

$$1. P_N(n) = (1-p)^{n-1} p \quad ; N \sim \text{Geol}(p) \quad , n=1,2,3,\dots$$

$$X_i \sim \text{Ber}(p) \quad M \sim \exp(\lambda)$$

$$a) X = X_1 + X_2 + \dots + X_N = \sum_{i=1}^N X_i$$

$$\begin{aligned} E[X] &= E\left[\sum_{i=1}^N X_i\right] = E\left[E\left[\sum_{i=1}^N X_i \mid N=n\right]\right] = E\left[\sum_{i=1}^n E[X_i \mid N=n]\right] \\ &= E\left[\sum_{i=1}^n E[X_i]\right] = E[N] E[X_i] = 1/p \cdot p = 1 \end{aligned}$$

$$\begin{aligned} V[X] &= V\left[\sum_{i=1}^N X_i\right] = E\left[\left(\sum_{i=1}^N X_i\right)^2\right] - E[X]^2 \\ &= E\left[\sum_{i=1}^N X_i^2 + 2 \sum_{1 \leq i < j \leq N} X_i X_j\right] - E[X]^2 = E[N] E[X^2] + 2 E\left[\frac{N(N-1)}{2} X_i X_j\right] - E[X]^2 \\ &= E[N] E[X^2] + E[(N^2 - N)] E[X_i] E[X_j] - E[X]^2 \\ &= \frac{1}{p} \cdot (p(1-p) + p^2) + \left[\left(\frac{1-p}{p^2} + \frac{1}{p^2}\right) - \frac{1}{p}\right] \cdot p^2 - 1 \\ &= 1-p + 1-p = 2(1-p) \end{aligned}$$

$$b) f_X(x) = \lambda e^{-\lambda x}, x \geq 0$$

$$M = M_1 + M_2 + \dots + M_X = \sum_{i=1}^X M_i, \quad E[M_i] = \frac{1}{\lambda}, \quad E[M_i^2] = V[M_i] + E[M_i]^2 = \frac{1}{\lambda^2} + \frac{1}{\lambda^2} = 2/\lambda^2$$

$$E[M] = E\left[E\left[\sum_{i=1}^X M_i \mid X\right]\right] = E\left[\sum_{i=1}^X E[M_i]\right] = E[X] \cdot E[M_i] = 1/\lambda$$

$$\begin{aligned} V[M] &= V\left[\sum_{i=1}^X M_i\right] = E[M^2] - E[M]^2 \\ &= E[X] E[M_i^2] + E[X^2 - X] E[M_i] E[M_i] - E[M]^2 \\ &= 2/\lambda^2 + \{2(1-p) + 1 - 1\} \frac{1}{\lambda^2} (1/\lambda)^2 = 1/\lambda^2 + 2(1-p) \frac{1}{\lambda^2} = \frac{3-2p}{\lambda^2} \end{aligned}$$

$$2. A=2 \quad W: \text{weather} \sim \text{Ber}(1/2)$$

$$S = A + W \quad N|W=1 \sim N(0, 1^2) \quad N|W=0 \sim N(0, 2^2)$$

$$a) f_S(s), \quad S = N+2$$

$$S|W=1 \sim N(2, 1^2), \quad S|W=0 \sim N(2, 2^2)$$

$$\begin{aligned} f_S(s) &= f_{S|W}(S|W=1)P(W=1) + f_{S|W}(S|W=0)P(W=0) \\ &= \frac{1}{2} \cdot \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(s-2)^2}{2}\right) + \frac{1}{2} \cdot \frac{1}{4\sqrt{2\pi}} \exp\left(-\frac{(s-2)^2}{8}\right) \end{aligned}$$

$$\begin{aligned} b) \int_1^3 f_S(s) ds &= \frac{1}{2} \int_1^3 \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(s-2)^2}{2}\right) ds + \frac{1}{2} \int_1^3 \frac{1}{4\sqrt{2\pi}} \exp\left(-\frac{(s-2)^2}{8}\right) ds \\ &= \frac{1}{2} \left(P(-1 \leq Z \leq 1) + P(-0.5 \leq Z \leq 0.5) \right) = \frac{1}{2} \cdot 2 \left(0.3413 + 0.1915 \right) \approx 0.5328 \end{aligned}$$

3. Gubner Chapter 4. Problem 1

$$f_X(x) = C \quad 0 \leq x \leq 7$$

$$\int_0^7 C dx = 1 \quad \therefore C = 1/7$$

$$\therefore \int_5^7 f_X(x) dx = \frac{2}{7} \quad P(\text{Alarm}) = (2/7)^3$$

4. Gubner Chapter 4. Problem 20

Student's t density with ν degrees of freedom

$$f_\nu(x) = \frac{(1 + \frac{x^2}{\nu})^{-(\nu+1)/2}}{\sqrt{\nu} B(1/2, \nu/2)}, \quad -\infty < x < \infty$$

$$B(1/2, \nu/2) = \int_0^\infty u^{-1/2} (1+u)^{-\frac{\nu+1}{2}} du$$

$$\int_{-\infty}^\infty f_\nu(x) dx = \int_{-\infty}^\infty \frac{1}{\sqrt{\nu} B(1/2, \nu/2)} (1 + \frac{x^2}{\nu})^{-\frac{\nu+1}{2}} dx$$

with $e^\theta = 1 + \frac{x^2}{\nu}$, $x^2 = (e^\theta - 1)\nu$, $x = \sqrt{(e^\theta - 1)\nu}$

$$\frac{dx}{\sqrt{\nu}} = e^{\theta/2} d\theta$$

4. Gulner Chapter 4, Problem 20.

$$f_V(x) = \frac{1}{\sqrt{V} B(1/2, 1/2)} \cdot (1 + \frac{x^2}{V})^{-\frac{V+1}{2}}$$

$$B(1/2, 1/2) = \int_0^\infty (1-e^\theta)^{-1/2} e^{-\theta/2} d\theta$$

$$\int_{-\infty}^\infty f_V(x) dx = 2 \int_0^\infty f_V(x) dx \quad \text{with } e^\theta = 1 + \frac{x^2}{V}, \quad x = \sqrt{(e^\theta - 1)V}$$

$$= 2 \cdot \frac{1}{\sqrt{V} B(1/2, 1/2)} \int_0^\infty (e^\theta)^{-\frac{V+1}{2}} \cdot \frac{\sqrt{V}}{2\sqrt{(e^\theta - 1)}} e^\theta d\theta \rightarrow e^\theta d\theta = \frac{2x}{V} dx \quad \therefore dx = \frac{V}{2x} e^\theta d\theta = \frac{V}{2\sqrt{(e^\theta - 1)}} e^\theta d\theta$$

$$= \frac{\sqrt{V}}{2\sqrt{(e^\theta - 1)}} e^\theta d\theta$$

$$= \frac{1}{B(1/2, 1/2)} \int_0^\infty e^{-\theta/2} e^{-\theta/2} \cdot e^\theta \cdot (e^\theta (1 - e^\theta))^{-1/2} d\theta$$

$$= \frac{1}{B(1/2, 1/2)} \int_0^\infty e^{-\theta/2} (1 - e^\theta)^{-1/2} d\theta = \frac{B(1/2, 1/2)}{B(1/2, 1/2)} = 1$$

5. Gulner Chapter 5, Problem 17

$$Z = X + Y, \quad X \sim \text{Ber}(p) \quad Y \sim N(0, 1)$$

$$f_Y(y) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(y-0)^2}{2}} = \frac{1}{\sqrt{2\pi}} \exp(-\frac{y^2}{2})$$

a) $F_{Z|X}(z|i)$ for $i = 0, 1$

$$i=0 \quad F_{Z|X}(z|0) = P(Z \leq z | X=0) = P(Y \leq z) = \Phi(z)$$

$$i=1 \quad F_{Z|X}(z|1) = P(Z \leq z | X=1) = P(Y+1 \leq z) = \Phi(z-1)$$

$$F_Z(z) = P(X=0) F_{Z|X}(z|0) + P(X=1) F_{Z|X}(z|1) = (1-p)\Phi(z) + p\Phi(z-1)$$

$$f_Z(z) = \frac{d}{dz} F_Z(z) = \frac{d}{dz} \{(1-p)\Phi(z) + p\Phi(z-1)\}$$

$$= (1-p)\phi(z) + p\phi(z-1)$$

b) $f_{Z|X}(z|1) = \phi(z-1), \quad f_{Z|X}(z|0) = \phi(z)$

$$f_{Z|X}(z|1)P(X=1) + f_{Z|X}(z|0)P(X=0) = f_Z(z)$$

6. Gubner Chapter 7, Problem 6.

$$F_{X,Y}(x,y) = \begin{cases} x - 1 - \frac{e^{-y} - e^{-xy}}{y}, & 1 \leq x \leq 2, y \geq 0 \\ 1 - \frac{e^{-y} - e^{-2y}}{y}, & x > 2, y \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

$$f_{X,Y}(x,y) = \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x,y) = \begin{cases} \frac{\partial}{\partial y} (1 - e^{-xy}) & 1 \leq x \leq 2, y > 0 \\ 0 & x > 2, y > 0 \end{cases}$$

$$\therefore f_{X,Y}(x,y) = xe^{-xy}, \quad 1 \leq x \leq 2, y > 0.$$

$$f_X(x) = \int_0^\infty xe^{-xy} dy = -\frac{1}{x} e^{-xy} \Big|_0^\infty = 1$$

$$f_Y(y) = \int_1^2 xe^{-xy} dx = -\frac{1}{y} xe^{-xy} - \frac{1}{y^2} e^{-xy} \Big|_1^2 = -\frac{2}{y} e^{-2y} - \frac{1}{y^2} e^{-2y} + \frac{1}{y} e^{-y} + \frac{1}{y^2} e^{-y}$$

$$\lim_{y \rightarrow \infty} F_{X,Y}(x,y) = F_X(x)$$

$$\therefore F_X(x) = \begin{cases} x-1 & 1 \leq x \leq 2 \\ 1 & x > 2 \end{cases}$$



7. Gubner Chapter 7, Problem 9

$$f_{X,Y}(x,y) = \frac{4e^{-(x-y)^2/2}}{y^5 \sqrt{2\pi}}, \quad y \geq 1$$

$$f_Y(y) = \int_{-\infty}^{\infty} \frac{4e^{-(x-y)^2/2}}{y^5 \sqrt{2\pi}} dx = \frac{4}{y^5 \sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{(x-y)^2}{2}} dx$$

$$\frac{(x-y)^2}{2} = t \quad \frac{2(x-y)}{2} dx = dt$$

$$\frac{1}{\sqrt{2t}} dx = dt$$